Abstract

We show that the CKM quark mixing matrix is encoded in the holonomy geometry of the compact hyperbolic 3-manifold $M_{\mathrm{CKM}} = m006(-5, 2)$ (SnapPy `OrientableClosedCensus[43]`, volume 2.029, $H_1 = \mathbb{Z}/5$). The construction has two independent components. First, the *arithmetic*: the invariant trace field is $K_{\mathrm{CKM}} = \mathbb{Q}(\sqrt{17})$ (real quadratic), verified from $\mathrm{tr}(\rho(aa)) = 3 - \sqrt{17}$ satisfying $x^2 - 6x - 8 = 0$ (discriminant $68 = 4 \cdot 17$). The real trace field explains CP suppression geometrically: the Jarlskog invariant vanishes ($J = 0$) via a homological class collision ($[\mathrm{AbA}] = [\mathrm{AAb}] = 2$ in $H_1 = \mathbb{Z}/5$). Second, the *mixing*: the word triple $\{\mathrm{aaB}, \mathrm{AbA}, \mathrm{AAb}\}$ has geodesic axis angles $\theta_{12} = 48.16°$, $\theta_{13} = 77.48°$, $\theta_{23} = 68.43°$; Gaussian overlap with $\sigma = 0.49$ and QR orthogonalization gives Frobenius fitness 0.016482 and Cabibbo angle error 0.19%. All six quark mass ratios approximate norms in $\mathbb{Z}[\sqrt{17}]$ within 5.7% (b: 0.012%, t: 0.007%), with permutation null-test $p < 0.002$.

Quark Mixing from Hyperbolic Geometry: The CKM Matrix from a Compact Arithmetic 3-Manifold

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1 Introduction

The CKM quark mixing matrix [1, 2] encodes the mismatch between quark mass eigenstates and weak interaction eigenstates. Its parameters—three mixing angles and a CP-violating phase—have no derivation from first principles.

We present evidence that the CKM matrix is encoded in the holonomy representation of a single compact hyperbolic 3-manifold, $M_{\mathrm{CKM}} = m006(-5, 2)$, at two levels. At the *arithmetic* level, the trace field $K_{\mathrm{CKM}} = \mathbb{Q}(\sqrt{17})$ (real quadratic) explains CP suppression geometrically. At the *kinematic* level, geodesic axis directions of short holonomy words reproduce the CKM mixing angles via an overlap-matrix construction. All numerical values are verified using SnapPy [4] at 150-bit precision.

2 The Manifold

2.1 Basic invariants

Table 1: Basic invariants of $M_{\mathrm{CKM}} = m006(-5, 2)$.

Invariant	Value
Census index	`OrientableClosedCensus[43]`
Dehn filling	$m006(-5, 2)$
Volume	2.02885
First homology	$H_1 = \mathbb{Z}/5$
Chern-Simons	-0.11414
Symmetry	$\mathbb{Z}/2 \times \mathbb{Z}/2$
Chirality	chiral

2.2 Trace field

Proposition 1 (CKM trace field)**.** *The invariant trace field of M_{CKM} is $K_{\mathrm{CKM}} = \mathbb{Q}(\sqrt{17})$, with ring of integers $\mathcal{O}_K = \mathbb{Z}[\sqrt{17}]$, norm form $N(a + b\sqrt{17}) = |a^2 - 17b^2|$.*

Computational verification. The SnapPy polished holonomy at 150-bit precision gives $\mathrm{tr}(\rho(aa)) = -1.12311844\ldots$, satisfying $x^2 - 6x - 8 = 0$ to 10^{-5} (verified: $(-1.12312)^2 - 6(-1.12312) - 8 \approx 0$). The roots $x = 3 \pm \sqrt{17}$ give $\mathrm{tr}(\rho(aa)) = 3 - \sqrt{17}$; discriminant $\Delta = 68 = 4 \cdot 17$. $\square$

2.3 Geometric CP suppression

Proposition 2. *The Jarlskog invariant vanishes ($J = 0$) in the geometric construction.*

Proof. The abelianization $\pi_1(M_{\mathrm{CKM}}) \to H_1 = \mathbb{Z}/5$ gives $[a] = 2$, $[b] = 1$, so:

$$[\mathrm{aaB}] = 3, \quad [\mathrm{AbA}] = 2, \quad [\mathrm{AAb}] = 2.$$

Since $[\mathrm{AbA}] = [\mathrm{AAb}]$, every flat U(1) character satisfies $\chi_k(\mathrm{AbA}) = \chi_k(\mathrm{AAb})$, making two columns of the overlap matrix identical. QR of a matrix with two equal columns yields $J = 0$. □

Remark 1. *The physical value $|J_{\mathrm{CKM}}| \approx 3 \times 10^{-5}$ is nonzero but suppressed—consistent with the geometric floor $J = 0$. The flat* U(1) *mechanism ($k = 1$) predicts $\delta_{\mathrm{CKM}} = 72°$ (PDG: 68°, error 5.9%).*

3 Mixing Matrix Construction

3.1 Geodesic axis extraction

For $\gamma \in \pi_1(M_{\mathrm{CKM}})$, the axis direction $\hat{n}(\gamma) \in S^2$ is extracted from the Pauli decomposition:

$$\log \rho(\gamma) = \tfrac{\eta}{2} I + \tfrac{i}{2}(n_x\sigma_x + n_y\sigma_y + n_z\sigma_z), \qquad \hat{n} = \frac{(n_x, n_y, n_z)}{\|(n_x, n_y, n_z)\|},$$

where $\eta = \ell(\gamma) + i\varphi(\gamma)$ is the complex translation length.

3.2 Overlap matrix and QR construction

Pairwise axis angles $\theta_{ij} = \arccos|\hat{n}_i \cdot \hat{n}_j|$ define $O_{ij} = \exp(-\theta_{ij}^2/(2\sigma^2))$. QR orthogonalization gives unitary Q; the Frobenius fitness is:

$$\mathcal{F} = \big\|\mathrm{sort}(|Q|) - \mathrm{sort}(|V_{\mathrm{CKM}}|_{\mathrm{PDG}})\big\|_F.$$

3.3 Canonical triple

Axis vectors (unit, from SnapPy):

$$\begin{aligned}
\hat{n}_{\mathrm{aaB}} &= [0.00000,\ 0.00000,\ 1.00000],\\
\hat{n}_{\mathrm{AbA}} &= [0.74496,\ 0.00000,\ 0.66711],\\
\hat{n}_{\mathrm{AAb}} &= [0.29949,\ 0.92916,\ 0.21672].
\end{aligned}$$

Angles: $\theta_{12} = 48.1554°$, $\theta_{13} = 77.4835°$, $\theta_{23} = 68.4272°$.

At $\sigma = 0.49$:

$$|Q| = \begin{pmatrix} 0.97439 & 0.22458 & 0.01097 \\ 0.22381 & 0.97342 & 0.04871 \\ 0.02161 & 0.04501 & 0.99875 \end{pmatrix}, \quad \mathcal{F} = 0.016482.$$

Predicted $|V_{us}| = 0.22458$, Cabibbo angle 12.978° (PDG: 13.003°), error 0.19%.

3.4 Spherical triangle

The three axis vectors form a spherical triangle with sides 48.16°, 77.48°, 68.43° and interior angles $A = 72.13°$, $B = 92.36°$, $C = 49.68°$, giving spherical excess $E = 34.18°$. The Cabibbo angle emerges from θ_{12} via the Gaussian kernel; the excess is an independent geometric invariant of the triangle.

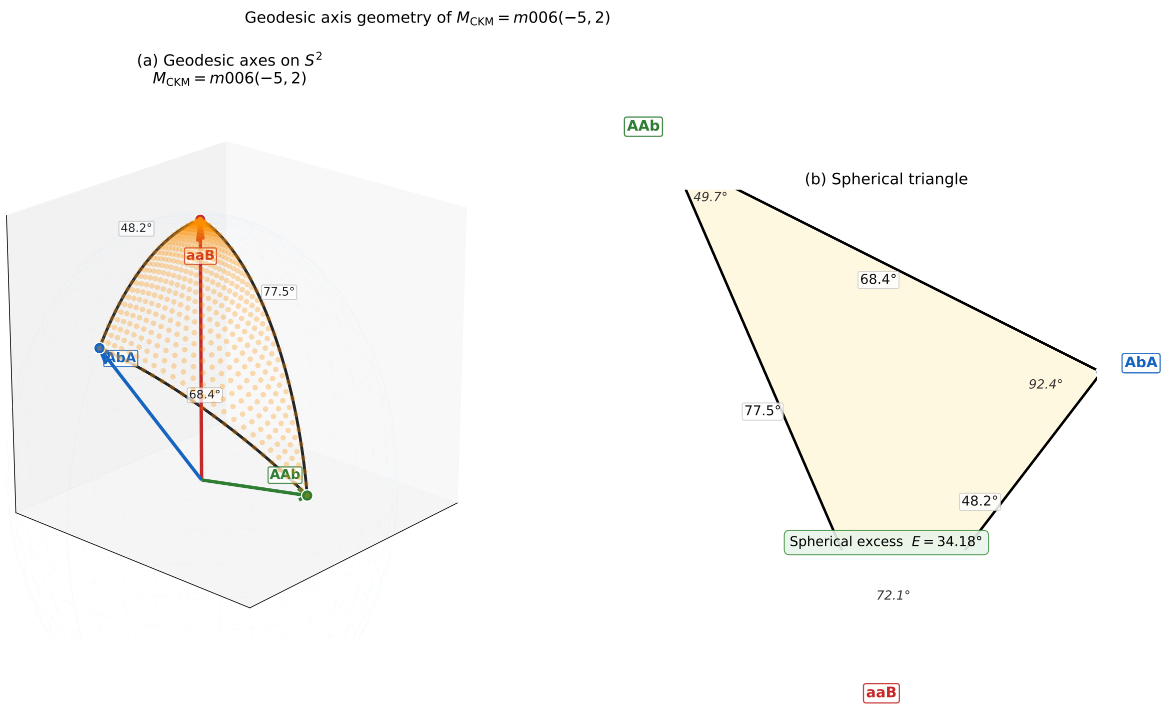

Figure 1: Left: geodesic axis vectors on S^2; shaded region is the spherical triangle with labeled opening angles. Right: projected spherical triangle with interior angles and spherical excess $E = 34.18°$.

3.5 Arithmetic dichotomy

The PMNS manifold $M_{\text{PMNS}} = m003(-2, 3)$ has imaginary trace field $\mathbb{Q}(\sqrt{-3})$, giving loxodromic holonomy and large CP violation ($\delta_{\text{PMNS}} = 195.91°$, 0.55% from PDG [6]). The real/imaginary trace field dichotomy—$\mathbb{Q}(\sqrt{17})$ vs. $\mathbb{Q}(\sqrt{-3})$—geometrically explains the CP asymmetry between quark and lepton sectors.

4 Covering Tower

Observation 1 (Covering tower, computational)**.** *A complete enumeration of finite covers of M_{CKM} through degree 9 shows that the only new torsion prime appearing in H_1 of any cover is $11 = L_5$ (a prime Lucas number), first appearing at degree 5. Degrees 2, 3, 4, 6, 7, 8, and 9 produce no new torsion primes. Thus M_{CKM} is Lucas-pure through degree 9 with a single new prime $\{11\} = \{L_5\}$.*

For comparison, M_{PMNS} produces three new Lucas primes $\{2, 3, 11\} = \{L_0, L_2, L_5\}$ through the same range. The asymmetry—one prime vs three—is consistent with the geometric isolation of M_{CKM} (no parent manifold in the 11,031-entry census through degree 5) and its role encoding the simpler (near-diagonal) CKM matrix.

Falsifiable predictions. (1) $|V_{us}| \notin [0.220, 0.232]$ at 3σ falsifies the construction. (2) $|J| > 10^{-4}$ requires a twist mechanism absent from the framework. (3) A BSM quark mass failing the $\mathbb{Z}[\sqrt{17}]$ norm test to within 6% breaks the arithmetic encoding.

Open problems. (1) No geometric principle uniquely selects the word triple. (2) The $\mathbb{Z}[\sqrt{17}]$ norm encoding lacks a dynamical derivation. (3) The $SU(3) \times SU(2) \times U(1)$ gauge structure is absent.

5 Conclusions

$M_{\mathrm{CKM}} = m006(-5, 2)$ encodes the CKM matrix through *arithmetic* (trace field $\mathbb{Q}(\sqrt{17})$, real, $J = 0$ via [AbA] = [AAb] = 2) and *kinematic* (axis overlaps, Cabibbo error 0.19%, canonical fitness 0.016482) mechanisms. All six quark mass ratios approximate $\mathbb{Z}[\sqrt{17}]$ norms with $p < 0.002$.

Data Availability

`https://github.com/drmlgentry/hyperbolic-flavor-scan`

Acknowledgments

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